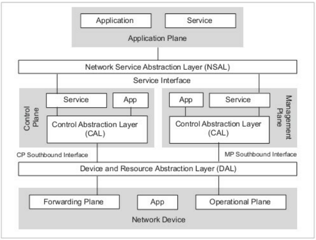


Figure 2: SDN planes

Let's take a closer look at the multiple planes of SDN.

**Forwarding plane:** This primarily performs the task of handling packets in the data path based on the instructions received from the control plane. It is the termination point for control-plane services and applications.

**Operational plane:** This is responsible for managing the operational state of the network devices, i.e., to observe whether the device is active/inactive, the number of ports on the device and the status of every port, etc. It is the termination point for the management plane's services and applications.

**Control plane:** This performs the duty of making decisions on how packets should be forwarded by network devices and focuses on the forwarding plane. It also fine-tunes the forwarding tables in the forwarding plane on the basis of network topology.

**Management plane:** This performs the tasks of monitoring, configuring and maintaining the network devices. It primarily focuses on the operational plane of the device rather than the forwarding plane. It also drafts all the forwarding rules for the network devices.

**Application plane:** All the applications and services define the application plane. It contains SDN applications for various functionalities such as network management, policy implementation and security services.

## Open source SDN platforms

### OpenDayLight

The OpenDayLight project is a collaborative SDN project initiative of the Linux Foundation. It uses open protocols to provide centralised, programmatic control and network device monitoring. OpenDayLight supports OpenFlow, as well as other ready-to-install network solutions.

It provides an interface to network administrators to connect to network devices efficiently and intelligently, for the best network performance.

It can be deployed in varied network environments and supports a modular controller framework as well as SDN standards and protocols. The OpenDayLight controller uses open northbound APIs, which are used by applications, which in turn use the controller to collect information about the network and create the rules for it, and run different algorithms to conduct analytics.

OpenDayLight can be deployed on hardware and operating system platforms that support Java.

***Features***

- Facilitates efficient control of devices with standard and open protocols.
- Provides centralised programmatic control of the physical and virtual devices in the network.
- Provides higher-level abstraction with tons of features to help network engineers create new applications and to customise network administration.
- Provides proactive support for network management and traffic flow.

***Official website:*** *https://www.opendaylight.org*
***Latest version:*** Oxygen

### OpenContrail

OpenContrail is a 2.0 licensed project that provides network virtualisation functionality on OpenStack and other orchestration systems. It provides all the necessary components for network virtualisation like the SDN controller, virtual router, analytics engine and northbound APIs. It includes extensive REST APIs to configure and gather data for operations and analytics from the system.

OpenContrail acts as a fundamental network platform for the cloud infrastructure.

The following are the components of the OpenContrail platform:

- **Controller:** This performs functions like accepting and converting orchestrator requests for VM creation, translating requests, and assigning networks. The real-time analytics engine collects, stores and analyses network elements; interacts with network elements for VM network provisioning and ensures uptime.
- **vRouter:** This virtualised routing element handles localised control plane and forwarding plane work on the compute node.
- **Gateway:** This eliminates the need for a software gateway and improves scalability and performance.

***Features***

- Service chaining: Routing of traffic based on predefined policy rules.
- Provides public IP addresses without NAT.
- Trunking: Provides support for all applications that